

Capstone Project 3

System Monitor Tool

Objective: Create a system monitor tool in C++ that displays real-time information about system processes, memory usage, and CPU load, similar to the 'top' command.

Day 1: Design UI layout and gather system data using system calls.

```
codebind@codebind:~/Desktop/project1$ gedit day1.cpp
Gtk-Message: 09:57:19.920: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.

(gedit:7812): GLib-GIO-WARNING **: 09:57:20.062: Error creating IO channel for /proc/self/mountinfo: Permission denied (g-file-error-quark, 2)
codebind@codebind:~/Desktop/project1$ g++ day1.cpp
codebind@codebind:~/Desktop/project1$ ./a.out
Total RAM In MB: 5907 MB
Total RAM In GB: 5 GB
Total unused RAM: 2506 MB
codebind@codebind:~/Desktop/project1$
```

Day 2: Display process list with CPU and memory usage.

```
codebind@codebind:~/Desktop/project1$ gedit day2.cpp
Gtk-Message: 10:00:28.486: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.

(gedit:7928): GLib-GIO-WARNING **: 10:00:28.644: Error creating IO channel for /proc/self/mountinfo: Permission denied (g-file-error-quark, 2)
codebind@codebind:~/Desktop/project1$ g++ day2.cpp
codebind@codebind:~/Desktop/project1$ ./a.out
User Time :17960
Nice Time :287
System Time :15425
Idle Time :5865179
iowait Time :1478
irq Time :0
softirq Time :194
steal Time :0
guest Time :0
guest_nice Time :0
idle_time:5866657
Total_CPU_Time:5900523
Active_Time :33866
CPU_Usage0
codebind@codebind:~/Desktop/project1$
```

Day 3: Implement process sorting by CPU and memory usage.

```
codebind@codebind:~/Desktop/project1$ gedit day3.cpp
Gtk-Message: 10:06:07.076: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.

(gedit:8049): GLib-GIO-WARNING **: 10:06:07.231: Error creating IO channel for /proc/self/mountinfo: Permission denied (g-file-error-quark, 2)

** (gedit:8049): WARNING **: 10:06:32.411: atk-bridge: get_device_events_reply: unknown signature
codebind@codebind:~/Desktop/project1$ g++ day3.cpp
codebind@codebind:~/Desktop/project1$ ./a.out
PID    CPU%    Memory (kB)    Name
8163    66.6667    0    (a.out)
1986    2.58887    0    (gnome-shell)
7651    0.920477    0    (gnome-terminal-)
4146    0.751994    0    (nautilus)
3555    0.203555    0    (gjs)
8157    0.165837    0    (tracker-extract)
4174    0.154555    0    (kworker/u16:5-events_unbound)
7669    0.109513    0    (kworker/0:0-events)
903     0.109124    0    (snapd)
1752    0.103613    0    (spice-vdagentd)
codebind@codebind:~/Desktop/project1$
```

Day 4: Add functionality to kill processes.

```
codebind@codebind:~/Desktop/project1$ gedit day4.cpp
Gtk-Message: 10:08:25.294: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.

(gedit:8166): GLib-GIO-WARNING **: 10:08:25.436: Error creating IO channel for /proc/self/mountinfo: Permission denied (g-file-error-quark, 2)
codebind@codebind:~/Desktop/project1$ g++ day4.cpp
codebind@codebind:~/Desktop/project1$ ./a.out
PID    CPU%    Memory (kB)    Name
1986    2.62067    0    (gnome-shell)
8278    2.5    0    (a.out)
7651    0.966308    0    (gnome-terminal-)
4146    0.73129    0    (nautilus)
8272    0.208333    0    (tracker-extract)
3555    0.200375    0    (gjs)
4174    0.15083    0    (kworker/u16:5-flush-252:0)
7669    0.116417    0    (kworker/0:0-events)
903     0.108328    0    (snapd)
1752    0.10312    0    (spice-vdagentd)
Enter pid to kill 8278
Killed
codebind@codebind:~/Desktop/project1$
```

Day 5: Implement real-time update feature to refresh data every few seconds.

```
PID      CPU%    Memory (kB)    Name
8590     11.1111 0              (a.out)
1986     2.80262 0              (gnome-shell)
8452     1.12137 0              (gnome-terminal-)
4146     0.689816 0              (nautilus)
3555     0.200921 0              (gjs)
4174     0.148811 0              (kworker/u16:5-events_power_efficient)
8135     0.143806 0              (kworker/0:1-events)
1752     0.111618 0              (spice-vdagentd)
7582     0.105892 0              (kworker/u16:0-events_power_efficient)
903      0.105512 0              (snapd)
Enter PID to Kill :
8590
Killed
codebind@codebind:~/Desktop/project1$
```